WHITE PAPER

Temporal Compression: Understanding Effective Storage Capacity

Physical capacity, processing time and retrieval in the assessment of temporal storage

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Abstract

Temporal storage uses compression processing to retain more original data within a fixed physical capacity. NSATS Temporal Compression makes compressed output available as input to a further compression process, connecting the amount of data retained with the work devoted to its representation. The executive question is whether that additional work reduces the infrastructure required to keep data available over its useful life. Answering it requires a distinction between physical capacity, effective capacity and processing rate, together with an account of retrieval. This paper develops those distinctions through a hypothetical example and examines their consequences for storage economics, environmental performance and the design of future services.

What temporal storage changes

A drive has a finite physical capacity, but the volume of original data it holds also depends on how that data is represented. When the same information can be stored using fewer bits, the drive can retain more of it. The physical resource remains the same; its effective capacity increases.

In NSATS's terminology, **temporal storage** uses compression processing to make this relationship part of the storage design. Processing takes time, and additional processing can produce a smaller stored form. **Temporal Compression** is the licensed technology. A **temporal drive** is a storage device that uses it.

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This meaning of temporal storage is distinct from keeping time-series measurements or historical versions of database records, and from temporal prediction in video compression, which uses relationships between pictures.

Time enters the decision twice: when data is prepared for storage and when it is retrieved. An organisation may be willing to spend more time storing an archive that will remain largely unchanged. It may make a different choice for a service whose records are updated and requested continuously. The value of a smaller stored form depends on that pattern of use.

Three capacity measures establish the vocabulary. **Physical capacity** is the capacity stated for the medium. **Usable capacity** is what remains after formatting, protection, system overheads and reserves. **Effective capacity** is the volume of original data that can be retained and returned intact, with the information needed for retrieval counted. Here, **intact retrieval** means returning data bit-for-bit identical to the original.

NSATS's stated basis for increased effective capacity is fewer stored bits per item of original data. This is a storage relationship, distinct from working RAM used temporarily during processing. Both resources matter to the system design, but they perform different roles.

NSATS's licensed Quantum technology enables one component to perform both storage and working-memory roles in a solution designed and built by the licensee, while existing drives and RAM remain ordinary classical storage. Where one component performs both roles, those roles and their demands on access remain distinct, and the capacity account treats them separately.

Effective capacity depends on processing time, not only physical size. A temporal drive is better described by how much data it can process and store per second than by a fixed capacity.

The practical specification therefore needs complementary measures: the amount retained, the rate at which it can be stored and the performance of retrieval. A processing rate describes the movement of data through the system. Retained capacity describes what remains available afterwards.

White Paper 1 introduces the wider NSATS portfolio; White Paper 2 examines information preservation through computation. This paper establishes the storage vocabulary and develops the investment question that follows from it.

Repeated compression and reversible computing

Lossless compression uses structure in data to represent it more compactly while preserving the information needed to return the original. Familiar examples include repetition and predictable relationships between values. A compressor must also retain enough information to interpret the result.

After an effective first pass, a second pass may find little remaining structure. Headers and other format information can then make the output larger. This is why recompressing an already compressed file often brings no benefit. Exceptions matter, however: for the special case of a file made entirely of zeros, zlib's authors note that the compressed output is likely to remain compressible and that another pass should produce further reduction. [1] Conventional methods also allow different amounts of processing effort in pursuit of a smaller output. [2]

The limit is general. There are fewer shorter binary files than files of a given length, so no lossless method can assign every possible input a distinct shorter encoding. Shannon's source-coding theory relates the average length of a lossless description to the source's information content. [3] Any information available separately to the decoder belongs in that account. A succession of passes is still a lossless transformation and remains subject to the same constraints.

The NSATS approach

NSATS licences enable classical devices to perform operations based on quantum mechanics, at room temperature.

Temporal Compression is developed technology available under licence. Its licensed software uses quantum-mechanical logic, and repeated compression is part of that technology. A compressed output can become the input to a further compression process. NSATS reports further reduction through these successive passes.

For storage, the licensee adds a chipset or controller built with the licensed technology. This provides the processing and control functions through which the data is stored and retrieved. The storage medium can be a hard drive, solid-state drive, USB flash drive, optical medium or an array.

NSATS sells technology licences, not chips. Its technology helps licensees manufacture their own chips.

A **pass** is one application of the compression process. The useful result of a further pass is the reduction in total retained size after the information needed for retrieval has been included. Processing effort, temporary memory and any intermediate writes are additional resource requirements. This makes repeated compression a decision about the complete storage task rather than the size of an isolated output file.

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What reversibility contributes

Reversibility concerns the preservation of information needed to identify an earlier state. Lossless compression needs an invertible relationship between the original data and its stored form. Establishing that relationship is different from establishing that every operation in the computing system is logically reversible, or that the total retained information has become smaller. Bennett's work demonstrates why time and storage requirements must be considered together in reversible computation. [4]

White Paper 2 describes NSATS's current arrangement. NSATS has achieved reversible computing at component level: its swap gates retain input and output information and work reversibly in a limited form. An external processor uses the retained information to regenerate earlier inputs. Full reversibility across the complete system is the subject of continuing research. In a storage implementation, any retained gate or processor information needed to retrieve the original belongs in the capacity account. Recoverability explains how earlier information remains available; the net stored size establishes the compression achieved.

A hypothetical drive example

Temporal Compression is operator-defined: the operator sets the amount of compression to suit its operational environment, with a minimum of 3%, measured as the reduction in stored size. The operator also defines the upper limit. The amount achieved above the minimum depends on the data, the drive, how the technology is used and where the licensed technology sits in the licensee's design.

The following example is hypothetical arithmetic, not an NSATS specification, typical figure or measured result. It applies the 3% minimum to 10 Gb of physical capacity and 10 Mb/s of original-data input. Stored size includes all supporting information; processing keeps pace with input; the ratio is constant; and no intermediate results are written to the medium. The simplified calculation makes no separate deduction for redundancy or reserved space.

The units are bits with decimal prefixes: **Gb** denotes gigabits and **Mb/s** megabits per second. Capital **B** denotes bytes. The byte equivalents are 10 Gb = 1.25 GB, 10.31 Gb ≈ 1.29 GB and 10 Mb/s = 1.25 MB/s.

Calculations involving compression use exactly the 3% minimum: stored size is 0.97 of original size, so the ratio is 1 ÷ 0.97, approximately 1.031. Derived figures are rounded only for display. A higher operator setting changes the ratio-dependent calculations; fixed inputs and baseline calculations remain unchanged, and whole-drive counts change only when a rounding threshold is crossed.

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Measure	Hypothetical value	Meaning
Physical capacity	10 Gb	Space provided by the medium
Compression ratio used	Approximately 1.031:1	Original size divided by stored size (1 ÷ 0.97)
Effective capacity at this ratio	Approximately 10.31 Gb	Original data retained within the physical capacity (10 Gb ÷ 0.97)
Stored size per 1 Gb of original data	0.97 Gb	1 − 0.03
Space saving	3%	The minimum operator setting
Original-data input rate	10 Mb/s	Incoming data before compression
Time to fill from empty at this ratio	Approximately 1,031 s, or about 17 minutes	(10 Gb ÷ 0.97) ÷ 10 Mb/s

The capacity gain and space saving differ because their denominators differ. The gain compares the additional original-data capacity with the storage resource. The saving compares the reduction with original size. A 3% saving and an approximately 3.1% gain describe the same result.

The time calculation makes a second distinction. Unprocessed input would fill the physical capacity in 1,000 s. At the assumed ratio, the drive holds more original data and takes longer to fill at the same original-data input rate. At the assumed input rate and ratio, a day of input, 864 Gb, would need 84 such drives, rounding 864 ÷ (10 ÷ 0.97) upwards. Continued processing therefore still requires additional storage if the whole stream is to be retained. A buffer is also needed whenever processing falls behind arrivals.

Retrieval is a separate operation. Neither its latency nor its throughput follows from the input rate or capacity calculations in this example.

From a smaller file to a usable storage system

The functional sequence is straightforward: accept the original data, perform the selected compression, retain the result and the information needed to read it, then use those retained resources to return the original. The implementation must preserve enough information to identify the necessary decoding steps, including any sequence of passes.

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The distinction between a file and a system matters here. A compressed file may depend on a dictionary, an index, processing metadata or retained state elsewhere. The Zstandard specification provides a conventional example: a compressed frame can refer to a dictionary supplied separately, which the decoder must obtain. [5] Excluding that dictionary from the file does not remove its storage cost or its role in retrieval.

Capacity accounting consequently includes the compressed data and its data-specific dependencies. Shared information is counted at system level. The decoding software is also a continuing dependency, although its size need not be charged separately to every file. Working memory is reported alongside persistent storage so that a smaller retained dataset is not mistaken for a reduction in every memory requirement.

Retrieval and continuity

For an interactive service, the delay before a requested object becomes available may be decisive. For a backup repository, the decisive question may be how quickly an entire service can be restored. These are different demands on processing, memory and storage access.

Access structure connects compression to retrieval. Independently decodable sections can reduce the work needed for a small request, but may limit the context available to the compressor. zlib's random-access guidance illustrates this trade-off. [6] A licensee must determine the smallest independently retrievable unit and the effect of simultaneous ingest and retrieval on its own design.

Intact retrieval can be checked against the original data, with cryptographic fingerprints such as SHA-256 providing a practical integrity check. [7] Durability adds a further requirement: the data and everything needed to read it must remain available through interruption, component replacement and the retention period. Damage to a compressed representation may affect a larger amount of original data, depending on its structure and dependencies. Protection therefore extends to decoding information as well as the compressed payload.

The OAIS archive model formalises this principle through representation information: preserving a digital object includes preserving the means of interpreting it. [8] For temporal storage, this connects software availability, controller dependencies and migration planning to the organisation's retention obligations. Moving to a system without the means to decode the stored form requires retrieval of the original data and provision for it at the destination.

Encryption and access

NSATS's software versions provide different encryption options or levels. The relevant distinctions concern authorised retrieval, protection of keys, processing overhead and the interaction with compression. Cryptographic strength is a separate property from compression ratio or reversibility.

Conventional systems commonly compress before encrypting. Special constructions can reverse that order: Johnson, Wagner and Ramchandran describe compression without access to the encryption key, with the necessary information available at decoding. [9] This establishes a scientific distinction, not NSATS's processing sequence.

For the storage operator, keys are part of the retrieval dependency. Their protection, authorised use and recovery arrangements have to last as long as the data, a concern treated separately in NIST's key-management guidance. [10]

When additional processing is commercially worthwhile

The commercial opportunity comes from the duration of the storage benefit. Processing occurs when data is ingested, recompressed or retrieved; a smaller stored form can reduce the capacity occupied throughout the period between those events. A collection retained for a long time and read infrequently can therefore present a different economic case from an equally large collection that changes continuously.

The investment comparison brings these costs onto a common basis. It includes drives, processing, controllers, working memory, licences, power, cooling and maintenance over the retention period. Migration and eventual replacement also belong in the comparison. The baseline is the incumbent system's actual footprint after its existing compression and deduplication.

An extension of the same hypothetical example shows the connection to equipment purchases. Assume a collection of 8,000 Tb and drives of 160 Tb each, equivalent to 1 PB on 20 TB drives. These are illustrative inputs. The ratio remains exactly 1 ÷ 0.97, and the calculation assumes no redundancy overhead.

Provisioning measure	Hypothetical result
Drives needed without the gain	50
Effective capacity per drive at the assumed ratio	Approximately 164.9 Tb
Drives needed at the assumed ratio	49, rounding 8,000 ÷ (160 ÷ 0.97) upwards
Drives not needed	1
Reduction in drive count	2%

The difference between space saved and drives avoided reflects purchasing in whole units. Where protection, reserves or spares reduce usable capacity, the same calculation starts from that lower capacity. A higher operator setting is calculated in the same way. The example gives an equipment count, not a financial or environmental saving.

For data centres, fewer required drives can defer purchases and make room for growth within existing racks. For a cloud provider charging on logical capacity, a smaller physical footprint can improve the economics of the service. Both organisations must also provision the processing and retrieval performance promised to users. A favourable storage result becomes commercially useful when that service can be delivered at a lower overall resource cost.

Workload variation remains part of provisioning. The behaviour of inputs that cannot be reduced, the capacity needed while processing is incomplete and the demand from simultaneous retrievals all affect the reserve required. Where decoding information is shared between customers, its protection and isolation are part of the service design.

Environmental consequences over the storage lifecycle

Fewer storage devices can reduce the manufacturing emissions [11] [12], materials and eventual waste associated with retaining a collection. Additional processors or controllers introduce their own resource requirements. The environmental comparison must therefore follow the equipment and the work performed across ingestion, retention and retrieval.

Electricity used by the IT equipment is only part of the facility total. Cooling and other overheads also matter, as the IEA's data-centre analysis explains. [13] Power usage effectiveness relates facility energy to IT equipment energy. [14] Water use requires a separate account reflecting the cooling arrangement and electricity supply. Where processing causes additional flash writes, endurance and replacement requirements also affect the result.

The useful comparison holds the retained data, protection, retrieval service and period constant. It then asks whether avoided storage hardware and operation outweigh the added processing and associated equipment. Energy savings and emissions reductions must remain distinct because the electricity source affects emissions. Greater efficiency per unit of data also need not reduce total consumption if it encourages a much larger retained collection.

This is why the hypothetical capacity and drive-count percentages cannot be treated as environmental percentages. Their significance is that they identify a possible change in equipment demand whose full consequences can then be measured.

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Applications and future storage services

The following applications are directions for licensees to assess. Their common question is how much processing a workload can accommodate in exchange for a smaller continuing storage requirement.

Retention according to access needs

An archive could allocate additional processing as records become less frequently accessed. Recently received or frequently requested material would remain on a path designed for prompt retrieval; older collections could receive further compression where its benefit justified the work. If access increased again, preparing the data for more frequent use would become part of the transition cost.

This suggests storage services differentiated by retrieval commitments as well as retained volume. A licensee could set policies for when to perform another pass, when to stop and how much processing to reserve for requests. Those policies would sit in its own design, with the effective-capacity target selected by the operator from the range of presented options.

Backup and industrial records

Backup repositories combine long retention with occasional urgent demand. Their case for additional compression depends on the gain beyond existing deduplication and on the time allowed for complete restoration. Capacity saved during retention has to be considered alongside the processing needed when a failure makes many records necessary at once.

Industrial recorders present a different constraint: collection may continue while earlier records are processed. A licensee could separate acquisition from background compression, provided staging capacity and processing keep the recording service within its limits. The design would be governed by arrival patterns, retention duration and the timing of later analysis.

Scientific and AI history

Simulation checkpoints, experiment versions and training records are candidates where retaining earlier states enables work to be reproduced or revisited. Additional compression after a checkpoint is recorded could reduce the cost of keeping that history. The actual formats and their existing encodings determine the opportunity; scientific and AI data should not be treated as uniformly compressible.

The retrieval task is as important as retention. Repeating a computation may require an entire dataset by a deadline, whereas an investigation may require a small selection from many versions. A useful design would preserve the associations between records and provide the access pattern needed for those tasks.

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Scheduling for electricity and processing availability

Optional passes could be scheduled during periods of spare computing capacity, lower electricity prices or lower electricity emissions. These objectives need not coincide. Published operational data has shown that flexible data-centre workloads can be scheduled around electricity emissions, providing a precedent for such a policy. [15]

A licensee would need to include the staging space used while data waits, any additional reads and writes, the deadline for completing storage and the energy required for later retrieval. Scheduling changes the timing of work; its value depends on the complete cost of that deferral.

Remote and constrained installations

Remote observatories, industrial edge systems and installations with limited space face a choice between retaining data locally and sending it elsewhere. Lossless compression is already established in space-data systems through CCSDS standards. [16] Temporal Compression offers a further storage approach for licensees to evaluate against that baseline.

Processing consumes power as well as time. Research on energy-aware compression shows why computation and memory costs have to be weighed against transmission costs. [17] A remote design would consequently consider local retention, connection availability, power supply and the means of retrieval at the destination together.

Specifying the service and making the decision

Temporal storage is best specified through the service it delivers. The starting quantities are original size, complete stored size, physical and usable capacity, and effective capacity. The accompanying time measures are processing time per pass and in total, sustained ingest throughput, retrieval latency and retrieval throughput. Peak resource use and concurrent activity explain the conditions under which those results can be achieved.

The specification also identifies persistent decoding information, temporary memory, integrity checks, protection and continuity dependencies. These details connect the capacity result to a service that can be maintained over the data's life. They allow a buyer to judge whether more data can be retained while meeting the organisation's access and restoration commitments.

NSATS licenses its intellectual property: patents, designs and copyrighted software. Where the licensed IP takes the form of software, NSATS supplies it under the licence, with its documentation and updates. Each licensee designs and builds its own solution.

Temporal Compression enables repeated processing of compressed output; the licensee determines how that capability serves a particular workload. A useful reduction in stored size must be considered over the period for which it persists, alongside the work needed to store and retrieve the data. Retained capacity, processing time and access requirements together establish the practical meaning of effective storage capacity.

Terms used in this paper

Term	Meaning
Temporal Compression	NSATS's licensed technology for compression using quantum-mechanical logic in classical devices, including repeated processing of compressed output.
Temporal drive	A storage device that uses Temporal Compression.
Temporal storage	A storage arrangement in which compression processing contributes to the effective capacity within a finite physical resource.
Physical capacity	The capacity stated for the storage medium.
Usable capacity	The capacity available after system overheads, protection and reserves are accounted for.
Effective capacity	The volume of original data retained and returned intact under stated conditions, with supporting information counted.
Original size	The size of the data before compression.
Stored size	The size of the retained representation and required supporting information within the stated boundary.
Working memory	Temporary storage used while processing; assessed separately from persistent information, including where one component performs both roles.
Compression ratio	Original size divided by stored size.
Space saving	The reduction expressed as a proportion of original size.
Effective-capacity gain	The increase in original-data capacity expressed relative to the storage capacity used as the baseline.
Pass	One application of a compression process.
Pass information	Information needed to identify and undo the sequence of processing, whether explicit or encoded within the representation.
Supporting information	Retained information required to interpret or retrieve the stored data.
Intact retrieval	Return of data bit-for-bit identical to the original.
Ingest throughput	Original data accepted and processed for storage per unit of time under stated conditions.
Retrieval latency	Delay before requested data becomes available.
Retrieval throughput	Original data returned per unit of time under stated conditions.

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The references establish scientific and operational background; the NSATS technology description and hypothetical assumptions are stated separately in the text.

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